

Experiment # 11

Configuring DHCP, DNS & HTTP Servers

Objectives

In this lab we will learn how to configure Web, DHCP and DNS Server.

- Understanding DHCP, DNS & HTTP
- How to configure DHCP, DNS & HTTP
- Configurations having DHCP within a LAN
- Configurations having DHCP outside the LAN

Understanding DHCP

DHCP is used by workstations (hosts) to get initial configuration information, such as an IP address, subnet mask, and default gateway upon boot up. Since each host needs an IP address to communicate in an IP network, DHCP eases the administrative burden of manually configuring each host with an IP address. Furthermore, if a host moves to a different IP subnet, it must use a different IP address than the one it previously used. DHCP takes care of this automatically. It allows the host to choose an IP address in the correct IP subnet.

DHCP Message Table

Message	Use
DHCPDISCOVER	The client is looking for available DHCP servers.
DHCPOFFER	The server response to the client DHCPDISCOVER.
DHCPREQUEST	The client broadcasts to the server, requesting offered parameters from one server specifically, as defined in the packet.
DHCPDECLINE	The client-to-server communication, indicating that the network address is already in use.
DHCPACK	The server-to-client communication with configuration parameters, including committed network address.
DHCPNAK	The server-to-client communication, refusing the request for configuration parameter.
DHCPRELEASE	The client-to-server communication, relinquishing network address and canceling remaining lease.
DHCPINFORM	The client-to-server communication, asking for only local configuration parameters that the client already has externally configured as an address.

Client-Server Conversation for Client Obtaining DHCP Address Where Client and DHCP Server Reside on Same Subnet

Packet Description	Source MAC Addr	Destination MAC Addr	Source IP Addr	Destination IP Addr
DHCPDISCOVER	Client	Broadcast	0.0.0.0	255.255.255.255
DHCPOFFER	DHCP Server	Broadcast	DHCP Server	255.255.255.255
DHCPREQUEST	Client	Broadcast	0.0.0.0	255.255.255.255
DHCPACK	DHCP Server	Broadcast	DHCP Server	255.255.255.255

How Dynamic Host Configuration Protocol (DHCP) Works?

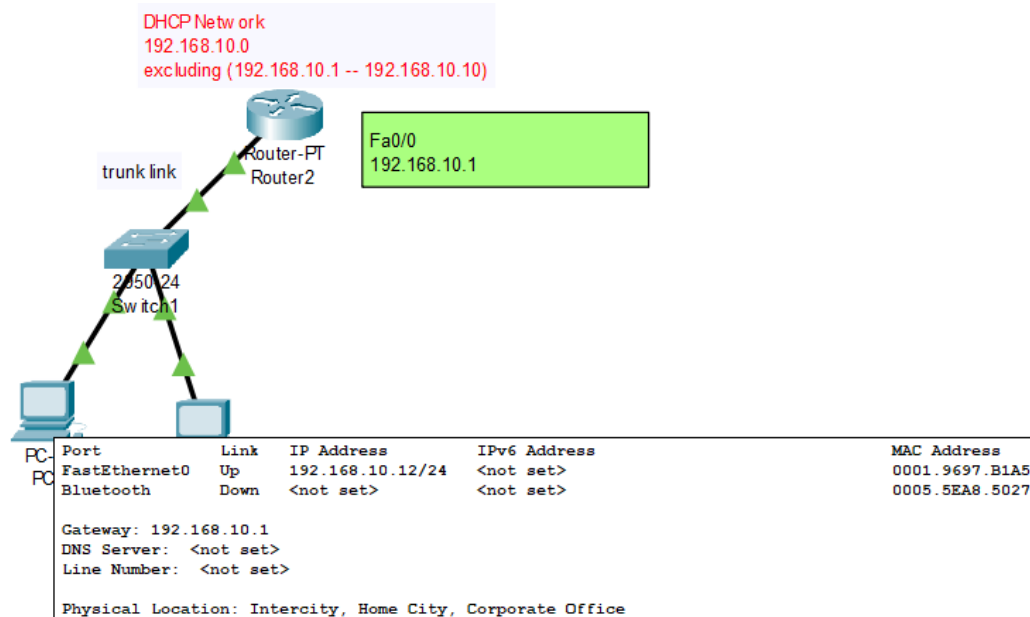
The Cisco **DHCP server** feature is a full implementation that **assigns and manages IP** addresses from specified address pools to DHCP clients. After a DHCP client has booted, the client begins sending packets to its default router. The IP address of the default router should be on the same subnet as the client.

The DHCP **discovery** follow **four steps**:

1. The client send a DHCPDISCOVER in broadcast to find a DHCP server.
2. The server sends a DHCPOFFER to the client in broadcast in case the DHCP server lies within a subnet otherwise in unicast/broadcast. In this phase the DHCP server sends a proposed configuration to the DHCP client.
3. The clients sends a DHCPREQUEST in broadcast, if the clients finds the offer agreeable. Why in broadcast? Because the DHCPDISCOVER send by the client might have reached more than one DHCP server.
4. The server sends a DHCPACK broadcast to the client.
 1. If the client detects that the ip is already in use it sends a DHCPDECLINE
 2. If the client receives a DHCPNAK from the server, it restart the process
 3. If there is no conflict with the ip address, the client can 'autoconfigure' the NIC.

ACTIVITY 1: Configuring DHCP on Router.

Place only the router and the switch of the following diagram on packet tracer and connect them.



Step 1: configure the fast Ethernet 0/0 port of router as:

IP address: 192.168.10.1 subnet mask: 255.255.255.0

And turn "Port Status" On.

Step 2: configuring DHCP on Router

Write the following in router configuration mode

```
Router0(config)# IP DHCP POOL pool1
Router0(dhcp-config)# network 192.168.10.0 255.255.255.0
Router0(dhcp-config)# default 192.168.10.1
Router0(dhcp-config)# exit
Router0(config)# ip dhcp exc 192.168.10.1 192.168.10.10
Router0(config)# do wr
Router0(config)# exit
```

Step 3: Attach the three PCs as shown above and turn the mode from static to dhcp in config tab of all these PCs.

We will see that router will automatically assign IP addresses and subnet masks to these PCs.

WHAT ARE DNS SERVERS?

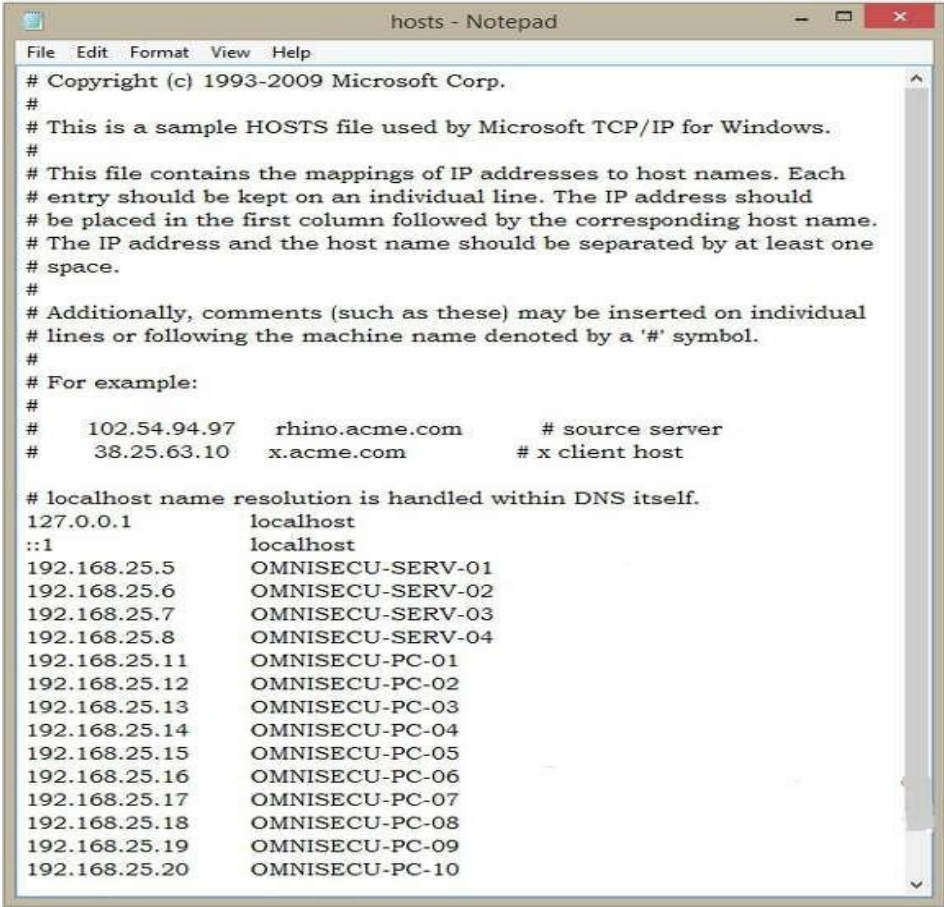
The networks which are running on TCP/IP protocol suite use IP Address ([IPv4 Address](#) or [IPv6 Address](#)), as the addressing mechanism. IP address ([IPv4 Address](#) or [IPv6 Address](#)) uniquely identifies a TCP/IP device in the network. However, it is difficult for the humans to remember IP Addresses for all the computers they want to communicate. Think about difficulty in remembering the IP Addresses of 1000s of web sites we use if we do not have the option to use their names. Humans prefer to use names instead of IP Addresses. Hence, we need a system to map human readable names to corresponding IP Addresses.

Text files called "hosts" files are used for name resolution in very early networks.

In Microsoft Windows Operating Systems, "hosts" file (without any file extension) is still in use (Windows 8.1). "hosts" file is located inside the folder "%WINDIR%\System32\drivers\etc". The host file has two columns. IP Address ([IPv4 Address](#) or [IPv6 Address](#)) and its corresponding name.

Note: %WINDIR% is a Windows Operating System Environment Variable, set to the location of Operating System folder. Copy the above folder path, paste it in your "Run" box and hit "OK" to open the folder.

Image of a "hosts" file used in Microsoft Windows Operating System is shown below.



```
File Edit Format View Help
# Copyright (c) 1993-2009 Microsoft Corp.
#
# This is a sample HOSTS file used by Microsoft TCP/IP for Windows.
#
# This file contains the mappings of IP addresses to host names. Each
# entry should be kept on an individual line. The IP address should
# be placed in the first column followed by the corresponding host name.
# The IP address and the host name should be separated by at least one
# space.
#
# Additionally, comments (such as these) may be inserted on individual
# lines or following the machine name denoted by a '#' symbol.
#
# For example:
#
# 102.54.94.97 rhino.acme.com # source server
# 38.25.63.10 x.acme.com # x client host

# localhost name resolution is handled within DNS itself.
127.0.0.1 localhost
::1 localhost
192.168.25.5 OMNISECU-SERV-01
192.168.25.6 OMNISECU-SERV-02
192.168.25.7 OMNISECU-SERV-03
192.168.25.8 OMNISECU-SERV-04
192.168.25.11 OMNISECU-PC-01
192.168.25.12 OMNISECU-PC-02
192.168.25.13 OMNISECU-PC-03
192.168.25.14 OMNISECU-PC-04
192.168.25.15 OMNISECU-PC-05
192.168.25.16 OMNISECU-PC-06
192.168.25.17 OMNISECU-PC-07
192.168.25.18 OMNISECU-PC-08
192.168.25.19 OMNISECU-PC-09
192.168.25.20 OMNISECU-PC-10
```

"hosts" files can be used for name resolution in small networks. But "hosts" file is not a solution for large networks like Internet. Is it possible to keep all the domain names in the Internet inside the "host" files in your computer? Impossible!

Domain Name System (DNS) is a solution to map human readable names to corresponding IP Addresses ([IPv4](#) or [IPv6](#)) and vice versa, in big networks. Domain Name System (DNS) was originally developed for name resolution in the early days of the Internet.

We (humans) use readable names to connect the Internet. Example: www.cisco.com, www.microsoft.com, www.google.com, www.redhat.com etc. All these names are resolved to their corresponding IP addresses, with the help of global DNS infrastructure. Most services offered by the Internet rely on Domain Name System (DNS) to work. Failure of global Domain Name System (DNS) infrastructure can also cause the failure of Internet (atleast for a short period of time, till we learn the corresponding IP Addresses of the names).

Instead of static "hosts" files, Domain Name System (DNS) is formed using a global infrastructure of DNS servers with a hierarchical structure of name resolution. Domain Name System (DNS) servers in the internet resolve only a segment of hosts, which they are

authorized to resolve. Hosts on the other segments are resolved by the authorized Domain Name System (DNS) servers of other segments by a concept called as delegation.

The global public Domain Name System (DNS) has a distributed architecture, serving from multiple DNS servers in multiple locations. This helps in load balancing (decreasing the work load on DNS servers) and redundancy (prevents single point of failure).

Hypertext Transfer Protocol (HTTP)

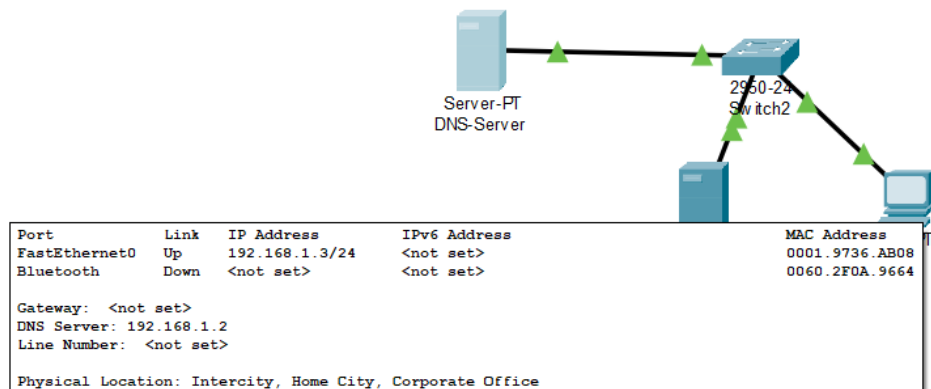
The operation of [Hypertext Transfer Protocol \(HTTP\)](#) involves the communication between a Hypertext Transfer Protocol (HTTP) client application (Usually web browser) and a Hypertext Transfer Protocol (HTTP) server application (Web servers like IIS). Hypertext Transfer Protocol (HTTP) uses [Transmission Control Protocol \(TCP\)](#) as the Transport Layer Protocol at [Well Known port number](#) 80. Once the [TCP](#) connection is established, the two steps in Hypertext Transfer Protocol (HTTP) communication are

- 1) **HTTP Client Request:** Hypertext Transfer Protocol (HTTP) client sends an Hypertext Transfer Protocol (HTTP) Request to the Hypertext Transfer Protocol (HTTP) Server according to the HTTP standard, specifying the information the client like to retrieve from the Hypertext Transfer Protocol (HTTP) Server.
- 2) **HTTP Server Response:** Once the Hypertext Transfer Protocol (HTTP) Request arrived at the Hypertext Transfer Protocol (HTTP) server, it will process the request and creates an Hypertext Transfer Protocol (HTTP) Response message. The Hypertext Transfer Protocol (HTTP) response message may contain the resource the Hypertext Transfer Protocol (HTTP) Client requested or information why the Hypertext Transfer Protocol (HTTP) request failed.



ACTIVITY 2: Configuring DNS & Web Server

Create this topology:



Step 1: IP configuration of web server as:

IP address: 192.168.1.5 subnet mask: 255.255.255.0

Step 2: IP configuration of DNS server as:

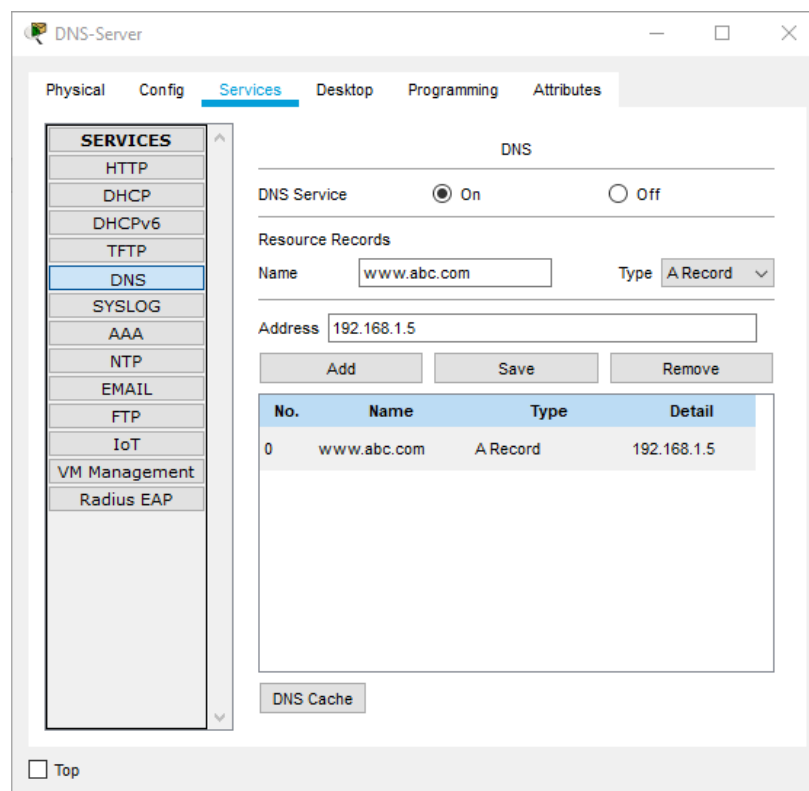
Ip address: 192.168.1.2
Subnet mask: 255.255.255.0
Default gateway: 192.168.1.1

Step 3: DNS configuration of DNS server.

Go to DNS Server -> services -> DNS and enter the following.

Name: www.abc.com

Address: 192.168.1.5 AND click “add”.



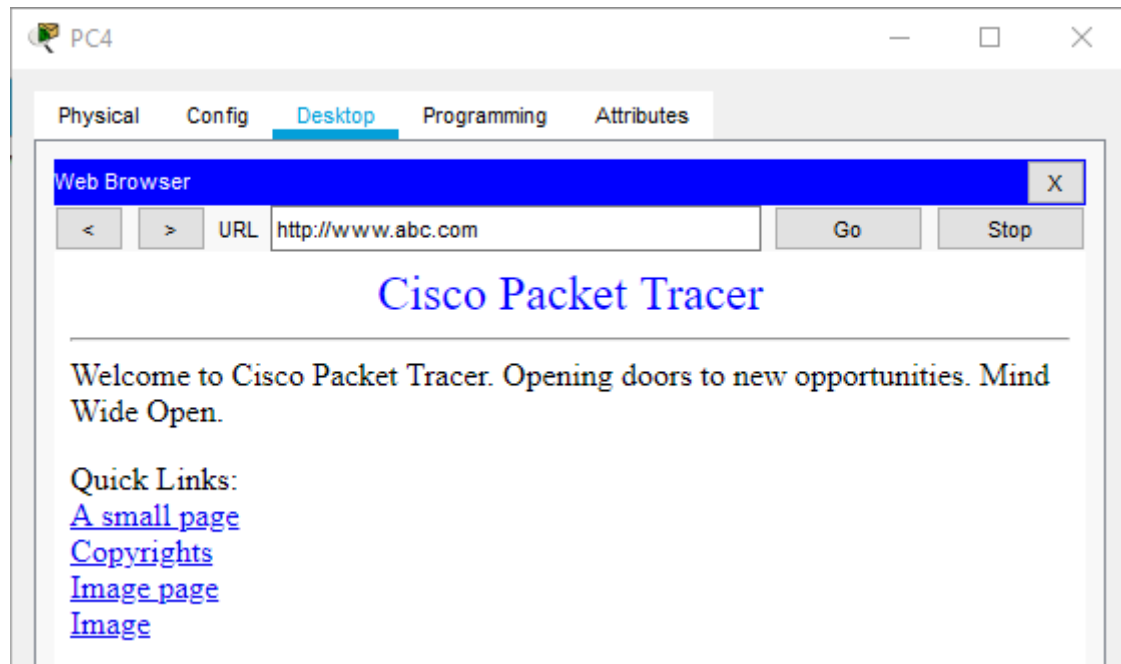
Step 4: IP configuration of End Device.

IP address: 192.168.1.3

Subnet mask: 255.255.255.0

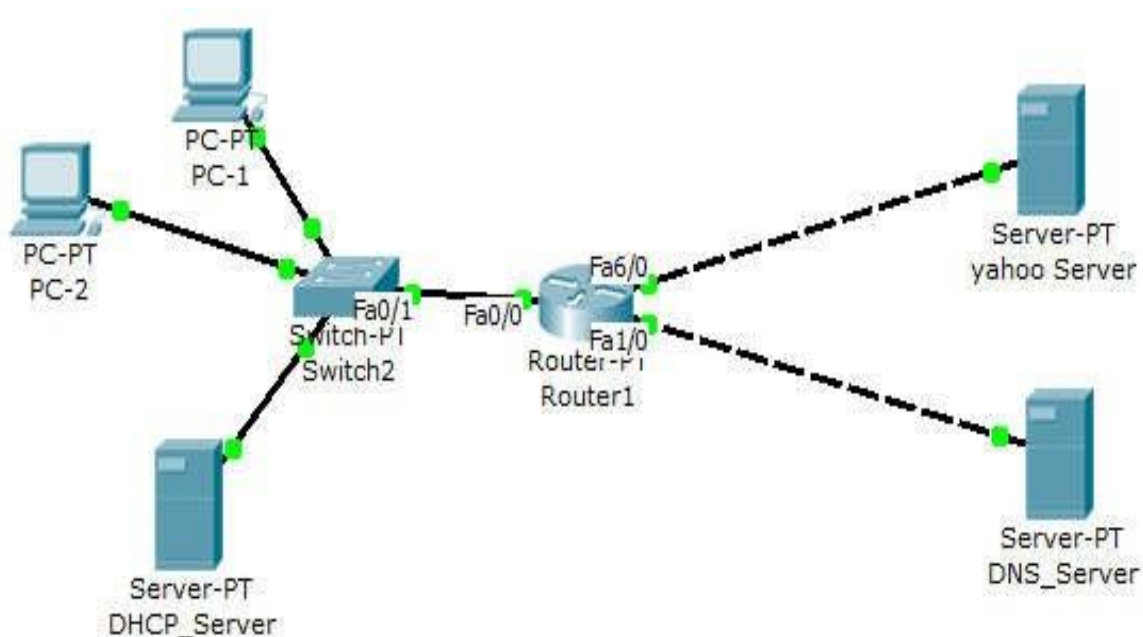
DNS Server: 192.168.1.2

Now go to End Device's Desktop interface, open web browser and enter www.abc.com in URL field. The html page is opened on your PC.



Activity 3: Configuring DHCP, DNS & HTTP Server having DHCP within the LAN.

Draw the following topology as shown:



Configure router ports i.e **fa0/0**, **fa1/0**, **fa6/0** as follows and turn “Port Status” ON.

Fast Ethernet 0/0

IP address: 192.168.1.1 Subnet mask: 255.255.255.0

Fast Ethernet 1/0

IP address: 192.168.2.1 Subnet mask: 255.255.255.0

Fast Ethernet 6/0

IP address: 192.168.3.1 Subnet mask: 255.255.255.0

➤ **IP configuration of DHCP server as**

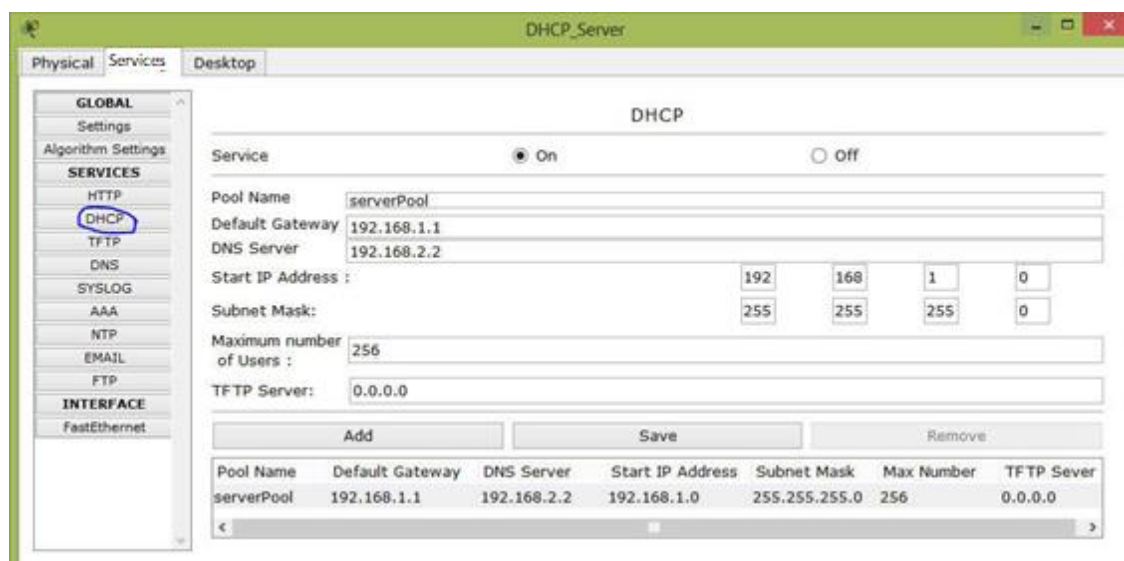
Ip address: 192.168.1.2

Subnet mask: 255.255.255.0

Default gateway: 192.168.1.1

Turn the PCs mode from static to dhcp in config tab of both of these PCs

Now DHCP configuration of DHCP server should be as follows



Keeping in mind that web server is connected at fa6/0 and DNS server is connected at fa1/0 do the following configurations.

➤ **IP configuration of Web Server**

IP address: 192.168.3.2

Subnet mask: 255.255.255.0

Default gateway: 192.168.3.1

DNS server Address: 192.168.2.2

➤ **DNS configuration of DNS server.**

Go to DNS Server -> services -> DNS and enter the following.

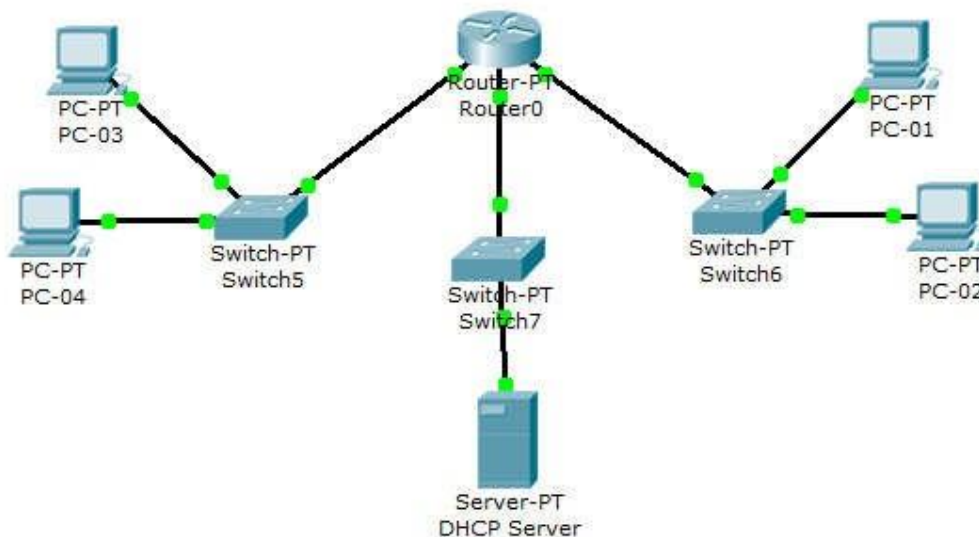
Name: www.yahoo.com

Address: 192.168.3.2 AND click “add”.

Now go to End Device’s Desktop interface, open web browser, and enter www.yahoo.com in URL field. The html page is opened on your PC.

Activity 4: Configuring DHCP outside the LAN.

Draw the following topology and configure router ports.



➤ Configure DHCP server as:

Ip address: 192.168.3.2

Subnet mask: 255.255.255.0

Default gateway: 192.168.3.1

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max Number	TFTP Server
serverPool	192.168.2.1	0.0.0.0	192.168.2.11	255.255.255.0	245	0.0.0.0

Similarly ADD second Pool with default Gateway 192.168.1.1

Now select router’s interfaces one by one and write the following command in third mode:-

Router0(config-if)# Ip helper-address 192.168.3.2

Lab Task

1. **Simulate** a configuration of DHCP server that will assign ip address of class B network.
2. Simulate a configuration of DNS server for your local area network. Assign different name to the PC and access the PCs by using the name instead of IP address.
3. Simulate a configuration of two web servers and access it by using IP address and by web address.

Conclusion
